

ABHINAY MESHRAM

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SUMMARY

Machine Learning Engineer with experience building end-to-end ML solutions using Python, Scikit-learn, FastAPI, Docker, and AWS EC2. Skilled in feature engineering, model optimization with Optuna, REST API development, and deploying production-ready machine learning applications.

SKILLS

Programming Languages: Python, SQL

Machine Learning: Supervised Learning(Classification & Regression), Feature Engineering, Data Preprocessing, Model Evaluation, Cross-Validation, Hyperparameter Tuning, Ensemble Learning, Imbalanced Learning (SMOTE)

Deep Learning: Artificial Neural Networks (ANN)

Data Analysis & Visualization: Exploratory Data Analysis, Statistical Analysis, Data Cleaning, Matplotlib, Seaborn

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, XGBoost, Optuna, FastAPI, TensorFlow, Keras

Deployment & APIs: REST APIs, FastAPI, Streamlit, Docker

Cloud & Tools: AWS EC2, Docker, Git, GitHub, Jupyter Notebook, Google Colab

PROJECTS

Credit Card Fraud Detection (Python, Scikit-learn, XGBoost, FastAPI, Streamlit) [Link](#) **Jun 2026**

- Built an end-to-end fraud detection system on **284,807** credit card transactions, performing EDA, feature scaling, SMOTE oversampling, and model comparison using Scikit-learn and XGBoost.
- Optimized a tuned **XGBoost** model using **RandomizedSearchCV** and **Stratified K-Fold Cross-Validation**, achieving **ROC-AUC 0.977**, **PR-AUC 0.869**, **93% precision**, and **82% recall**.
- Developed a **FastAPI REST API** and integrated it with a **Streamlit** dashboard for real-time and batch fraud prediction, probability scoring, threshold tuning, and interactive visualizations.

Customer Churn Prediction API (Python, Scikit-learn, Optuna, FastAPI, Docker, AWS EC2) [Link](#) **Jul 2026**

- Built an end-to-end Machine Learning pipeline for telecom customer churn prediction using feature engineering, **ColumnTransformer**, and Scikit-learn **Pipeline**, evaluating multiple classification models including Logistic Regression, Decision Tree, KNN, Random Forest, Gradient Boosting, AdaBoost, and XGBoost.
- Optimized the final **Random Forest** model using **Optuna** for hyperparameter tuning, achieving **Accuracy 0.77**, **Precision 0.57**, **Recall 0.56**, **F1 Score 0.57**, and **ROC-AUC 0.81**.
- Developed and deployed a production-ready **FastAPI REST API**, containerized the application using **Docker**, and deployed it on **AWS EC2**, enabling real-time predictions through an interactive Swagger UI.

Laptop Price Prediction (Python, Scikit-learn, Streamlit) [Link](#) **Mar 2026**

- Built an end-to-end **Machine Learning** pipeline for Laptop Price Prediction using Python and Scikit-learn on a dataset of 100,000 records with 18 hardware features, performing **exploratory data analysis (EDA)** with Matplotlib and Seaborn to identify key price-driving features.
- Performed **data cleaning**, **feature engineering**, and categorical encoding, then trained and validated Linear Regression, Decision Tree, Random Forest, and Gradient Boosting models using **k-fold cross-validation** to ensure robust generalization and reduce overfitting.
- Achieved an **R² score of 0.91** with the best-performing model and deployed a **production-ready**, interactive Streamlit web application for **real-time** price prediction.

WORK EXPERIENCE (INTERNSHIPS)

Tata – Data Visualisation: Empowering Business with Effective Insights **Sep 2025**

- Cleaned and preprocessed datasets performing exploratory data analysis to surface actionable trends
- Built **10+ interactive visualizations** (charts/dashboards) to support data-driven decision-making for stakeholders

Deloitte Australia – Data Analytics Job Simulation **Aug 2025**

- Analyzed structured datasets in Excel to identify business insights, applying data validation and quality checks across all records
- Performed data classification and quality assessment on large datasets to ensure analysis-ready accuracy

CERTIFICATIONS

- Machine Learning Mastery
- Data Science & Analytics with Artificial Intelligence
- Data Analysis with Python

EDUCATION

Bachelor of Science in Information Technology

University of Mumbai, Mumbai, India

Jun 2022 – Apr 2025

CGPA: 8.22 / 10